Qwen

Forging a Sovereign Coding Agent: A Self-Validating Framework of Machine-Readable Invariants

This research report outlines the design of a dual-purpose framework for the "Code-Command" sovereign coding agent. The framework is designed to establish a precise contract of invariants through ten machine-readable tags, which are then enforced by a minimal, custom-built validation system. This system operates entirely within a zero-setup GitHub and browser environment, adhering strictly to the use of only Rust, JavaScript, C++, and ALN/Markdown. The objective is to create a unified policy surface that guides both human and AI interactions while enabling robust, automated enforcement of the project's core principles.

Defining the Policy Contract: Ten Atomic Invariant Tags

The foundational element of the framework is a set of ten atomic, abbreviated, and machine-readable tags. These tags, prefixed with `CC-`, translate the high-level `(R)` rules of the Code-Command project into a formal, verifiable language. This approach creates a single policy surface that is simultaneously interpretable by humans providing instructions and by automated systems like AI chats and GitHub Actions [84] [144]. Each tag corresponds to a specific invariant derived directly from the user's directives, ensuring clarity and precision. The table below details each proposed tag, its definition based on the `(R)` rules, and the corresponding machine-checkable logic.

CC Tag ID	Definition & Invariant Logic	Aligned (R) Rule
CC-VOL	Volume Lock. Asserts that a response or artifact contains more than N distinct functions or automation blocks. It fails if the content is truncated, placeholder text, or otherwise lacks sufficient volume.	(R): Each response must yield a high-volume of code-functions, task-behavior, and automations...
CC-LANG	Sovereign Stack. Restricts file extensions and syntax to the explicitly supported languages: `.rs`, `.js`, `.cpp`, `.h`, `.aln`, and `.md`. It fails if any other extension, such as `.py` or `.ts`, is present.	(R): Only use the supported-languages (Rust, Javascript, CPP, and *ALN, or Markdown for Policies*)
CC-CRATE	Fresh Logic. Requires that every commit or delta introduces new declarations of `struct`, `impl`, `fn`, or `mod` that did not exist in a prior state or tree-sitter parse of the codebase.	(R): New crates, function-structs, configs, and logic must-be built in every conversation at some point.
CC-FILE	Destination Strict. Enforces the `[FILENAME]:[PATH]` pattern in a header comment preceding a code block. It fails if the code block lacks a preceding directive like `// FILE: ./path/name.ext`.	(R): Always include a filename, and destination for any code, snippet, file, document, etc.
CC-FULL	No Excerpts. Fails if a code block contains ellipsis (`...`), `// rest of code`, or `/<em> omitted </em>/`, thereby enforcing complete code generation.	(R): Do-not respond-with excerpts, samples, or illustrations.
CC-DEEP	Depth-3 Struct. Validates that directory navigation strings for certain core modules have a minimum depth of three path components (e.g., `./src/core/agents/mod.rs`).	(R): Three-level deep-directories are-required for some files, or javascript-functions...
CC-ZERO	No-Init Boot. Ensures that entry points like `main.rs` contain only relative path imports and zero references to setup functions, environment variables like `std::env::temp_dir`, or installation logic.	(R): Runs right-from the Github-repository without any setup required...
CC-PATH	Path Integrity. Scans path strings for malformed patterns, such as double slashes (`./path//file`) or OS-specific backslashes (`C:\\\\folder`), and fails on their detection.	(R): Paths must-not be malformed, misleading, spelled-incorrectly, or contain merged-instances of filenames...
CC-SOV	No Externals. Scans import/include statements (`use`, `require`, `#include`) for forbidden dependencies. It fails if it detects any crate or module not explicitly written by the agent itself.	(R): You do-not use, or mention external-sources, or external-tools: We build custom-tools, and custom-code for all functionality...
CC-NAV	Discovery Logic. Requires the presence of a custom-implemented Rust or C++ function containing logic for directory traversal, such as a recursive `PathBuf` iterator, to satisfy the need for pathfinding utilities.	(R): Uncover research-objects, pathfinding-logic, file-placement strategies...

These tags provide a structured and enforceable vocabulary for the agent's behavior. They are designed to be embedded within generated artifacts, for instance, in a YAML frontmatter block at the top of a code file, making them immediately checkable before further processing [56] [116].

The Architecture of Sovereignty: A Zero-Setup Browser-Based System

The enforcement of the `CC-` tags is realized through a sovereign architectural model that eliminates the need for servers, terminals, or external services. This architecture is predicated on running the entire Code-Command agent directly within a web browser, instantiated from a standard GitHub repository. This design choice is central to achieving the "zero-setup" principle and full dependency sovereignty. The core of this architecture is the compilation of the agent's logic—primarily written in Rust and C++—into a WebAssembly (WASM) binary [5] [40]. This WASM module acts as the agent's brain, executing all custom logic, including the invariant validation, entirely on the client side. This approach aligns with modern trends of agentic systems operating in terminal-native environments [41].

The interaction between the user and the agent is mediated through a browser-based frontend, likely an `index.html` file that serves as the single entry point for the application [46]. This HTML file would load a browser-based code editor component, such as CodeMirror, and a JavaScript runtime that instantiates the WASM module [97]. Communication between the JavaScript frontend and the Rust/C++ backend is managed through a minimal, custom-defined API, exposing specific functions like `cc_invoke` or `cc_validate_invariants` [70] [99]. This direct integration ensures that all orchestration and data flow remain under the project's control, avoiding reliance on external frameworks or services.

To fulfill its role as a coding agent, the system treats the GitHub repository as its sovereign filesystem. Instead of requiring local installation, the agent mounts the repository by making direct calls to the GitHub REST API [15]. Key endpoints are used for this purpose: the Git Trees API (`GET /repos/{owner}/{repo}/git/trees/{tree_sha}?recursive=1`) allows for the efficient, single-request retrieval of the entire repository structure, which is critical for implementing path-finding and navigation-related invariants like `CC-NAV` [11] [18]. Individual file contents are fetched using the Contents API (`GET /repos/{owner}/{repo}/contents/{path}`), with Base64-encoded content being decoded into usable text format within the browser [56]. All modifications made by the agent, such as generating new files or refactoring existing ones, are persisted back to the repository via the same API, using endpoints like `POST /repos/{owner}/{repo}/contents/{path}` with appropriate commit messages and file SHAs [173]. This entire workflow—from fetching the initial repository state to executing tasks and committing changes—is contained within the browser, making the agent truly

portable and instantly loadable from any valid GitHub repository URL. Client-side caching mechanisms, potentially using IndexedDB, can be implemented to store file content and metadata, reducing API calls, mitigating rate limits, and improving performance .

Tiered Enforcement: A Custom Validator Engine Without External Dependencies

The enforcement of the `CC-` invariant tags is carried out by a lightweight, custom-built validation engine. This engine is a direct consequence of the project's strict constraint against external tools and sources, which prohibits the use of standard parser libraries like `tree-sitter` or Rust's `syn` crate [1] [6]. To comply with this rule, the validation logic is implemented as a "Custom Token Walker," a hand-written state machine that scans code files as raw strings using only the standard libraries available in Rust or JavaScript [77] [130]. This approach provides the necessary structural awareness without introducing external dependencies, thus upholding the project's sovereignty.

The validation engine is structured in two tiers, allowing for a scalable and efficient enforcement mechanism. **Tier 1** handles invariants that can be checked with simple pattern matching and string scanning. Tags such as `CC-FILE`, `CC-FULL`, `CC-LANG`, and `CC-PATH` fall into this category. For example, validating `CC-FILE` involves a straightforward scan for the `// FILE:` prefix within the first few lines of a code block. Validating `CC-PATH` requires checking for invalid sequences like `//` or `\` within path strings. These checks can be efficiently implemented using basic string manipulation methods and regular expressions available in the standard library, ensuring they are fast and lightweight [90] [180]. The following Rust code snippet illustrates how a custom walker could validate path integrity without any external crates:

```
// FILE: ./core/invariants/src/custom_token_walker.rs
pub struct CCTokenWalker {
    content: String,
}

impl CCTokenWalker {
    pub fn check_path_integrity(&self) -> bool {
        let bytes = self.content.as_bytes();
        for i in 0..bytes.len().saturating_sub(1) {
```

```
            // Fail on double slash or Windows backslash
            if (bytes[i] == b'/' && bytes[i + 1] == b'/') || bytes[i] == b
                return false;
            }
        }
        true
    }
}
```

Tier 2 addresses more complex invariants that require lightweight structural analysis, such as `CC-CRATE` (detecting new logic) and `CC-NAV` (verifying navigation utilities). While a full compiler-grade Abstract Syntax Tree (AST) is not feasible, a simplified, heuristic-based analysis is possible [139][151]. An Abstract Syntax Tree (AST) is a conceptual representation of code structure, a fundamental CS concept, not an external tool . The validator can implement a minimal, custom scanner that identifies key syntactic constructs like function definitions (`fn`), struct declarations (`struct`), or recursive function calls. For `CC-CRATE`, the walker would scan for the appearance of new `fn` or `struct` keywords in a code delta compared to the previous version. For `CC-NAV`, it would look for a function signature containing logic indicative of recursive directory traversal. This "heuristic AST analysis" provides the necessary structural context to enforce the invariants accurately without resorting to external parsing libraries. This two-tiered, custom-built framework fully satisfies the project's requirements, offering a robust, self-contained, and sovereign method for validating AI-generated code.

Strategic Synthesis: Integrating Tags and Validation into a Unified Policy Surface

The successful implementation of the Code-Command framework hinges on the seamless integration of the ten invariant tags and the custom validation engine into a cohesive, sovereign architecture. This synthesis creates a unified policy surface where the `CC-` tags act as declarative guardrails, and the custom validator enforces them automatically, fulfilling the project's primary goals of guiding AI behavior and ensuring compliance without external dependencies. The process begins with the generation of code artifacts, which must include the `cc-invariants` list within a YAML frontmatter block, clearly stating the contractual obligations for that piece of code [56][57]. This declaration serves a dual purpose: it guides the AI during generation by specifying required attributes like `CC-FILE` and `CC-FULL`, and it provides the machine-readable input for the validator.

Upon generation, the code artifact is passed to the browser-based system. The frontend JavaScript captures the code and the associated tags, then invokes a pre-compiled WASM function exposed by the `CC-Engine` [99]. Inside the WASM module, the custom validator executes its tiered logic. For each tag in the provided list, it triggers the corresponding validation function. If any check fails—for instance, if `CC-SOV` detects a forbidden external import—the validator returns a failure status and an error message to the frontend. The frontend can then display this information to the user, preventing the submission of non-compliant code, or it can be integrated into a GitHub Action to block a pull request merge [61]. This entire feedback loop occurs locally in the browser, ensuring the system remains lightweight and independent of any server-side infrastructure.

Ultimately, this framework elevates the project beyond a simple coding utility by establishing a clear, enforceable, and self-contained governance model. By codifying its rules into machine-readable tags and building a sovereign validator to enforce them, Code-Command creates a resilient and portable environment for AI-assisted development. The design aligns with emerging concepts in agentic AI, such as "Policy-as-Prompt" and "Skills" that carry machine-readable policy labels, demonstrating a sophisticated approach to managing autonomous agent behavior [79] [101]. The resulting system is not just a tool but a secure, auditable, and highly portable platform for generating code that adheres to its own internal laws, perfectly embodying the principles of sovereignty and zero-setup.

Reference

1. How to parse Rust code without using procedural macros? https://stackoverflow.com/questions/57408862/how-to-parse-rust-code-without-using-procedural-macros
2. Compiling an AST back to source code - Stack Overflow https://stackoverflow.com/questions/5832412/compiling-an-ast-back-to-source-code
3. Security Considerations for Artificial Intelligence Agents (Perplexity ... https://arxiv.org/html/2603.12230v1
4. Designing Constraints for AI Code Generation | Sungat Arynov ... https://www.linkedin.com/posts/sungat-arynov_3-am-claude-code-hit-its-limit-800-lines-activity-7422175962504527872-bYuT
5. Compiling from Rust to WebAssembly - MDN Web Docs https://developer.mozilla.org/en-US/docs/WebAssembly/Guides/Rust_to_Wasm

6. How to enforce dependency rules in javascript/typescript/webpack https://stackoverflow.com/questions/51988012/how-to-enforce-dependency-rules-in-javascript-typescript-webpack
7. Reusing Legacy Code in WebAssembly: Key Challenges of Cross ... https://arxiv.org/html/2412.20258v1
8. Simple Index https://mirrors.aliyun.com/pypi/simple/
9. A Review of AI-Driven Automation Technologies: Latest Taxonomies ... https://www.sciopen.com/article/10.32604/cmc.2025.067857
10. zoom-oss-attribution-march-2024.txt https://www.zoom.com/siteassets/opensource/en/2024/zoom-oss-attribution-march-2024.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669
11. [PDF] ETOM: A Five-Level Benchmark for Evaluating Tool Orchestration ... https://aclanthology.org/2026.findings-eacl.75.pdf
12. Arxiv今日论文| 2026-04-09 - 闲记算法 http://lonepatient.top/2026/04/09/arxiv_papers_2026-04-09.html
13. Fine-tuning LLMs to Synthesize Invariants for Smart Contract Security https://arxiv.org/html/2510.21401v1
14. Findings of the Association for Computational Linguistics: EMNLP ... https://aclanthology.org/volumes/2025.findings-emnlp/
15. Dive into Claude Code: The Design Space of Today's and Future AI ... https://arxiv.org/html/2604.14228v1
16. Verify AI Code Trust in HIPAA Environments with TrustRun - LinkedIn https://www.linkedin.com/posts/nicholas-tatonetti_github-tatonetti-labtrustrun-network-activity-7428475478132535296-4o7D
17. Why does my GitHub page not update its content? - Stack Overflow https://stackoverflow.com/questions/24713112/why-does-my-github-page-not-update-its-content
18. zoom-oss-attribution-february-2024.txt https://www.zoom.com/siteassets/opensource/en/2024/zoom-oss-attribution-february-2024.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669
19. CWM: An Open-Weights LLM for Research on Code Generation with ... https://arxiv.org/html/2510.02387v1

20. AWS Security Hub Integration for Elastic https://www.elastic.co/docs/reference/integrations/aws_securityhub
21. [DOC] VS_ThirdPartyNotices.docx - Visual Studio - Microsoft https://visualstudio.microsoft.com/wp-content/uploads/2019/05/VS_ThirdPartyNotices.docx
22. Fixing Web Dependencies in the JS Ecosystem | Lea Verou, Ph.D ... https://www.linkedin.com/posts/leaverou_web-dependencies-are-broken-can-we-fix-them-activity-7414754650551291905-vvJS
23. Linux agent install checklist | FortiCNAPP - Fortinet Document Library https://docs.fortinet.com/document/forticnapp/26.2.0/administration-guide/376786/linux-agent-install-checklist
24. #ai #vscode #githubcopilot #developerexperience #agenticai ... https://www.linkedin.com/posts/jonathanvella_ai-vscode-githubcopilot-activity-7435198256877453312-3w-h
25. Chrome DevTools, Antigravity, Context7 & GitHub MCP highlights https://www.linkedin.com/posts/andrey-pogodin-75512_three-weeks-have-passed-since-i-shared-my-activity-7405889656636149761-6Y-C
26. Azure updates https://azure.microsoft.com/updates?id=558178
27. Static-analysis with custom rules for JavaScript? - Stack Overflow https://stackoverflow.com/questions/11816713/static-analysis-with-custom-rules-for-javascript
28. [PDF] Study of JavaScript Static Analysis Tools for Vulnerability Detection ... https://arxiv.org/pdf/2301.05097
29. eslint-plugin-import-x - NPM https://www.npmjs.com/package/eslint-plugin-import-x
30. JavaScript modules - MDN Web Docs - Mozilla https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Modules
31. eslint-plugin-import - NPM https://www.npmjs.com/package/eslint-plugin-import
32. Tree Shaking - webpack https://webpack.js.org/guides/tree-shaking/
33. WebAssembly.instantiate(): param count of 1055 exceeds internal ... https://stackoverflow.com/questions/72531744/compileerror-webassembly-instantiate-param-count-of-1055-exceeds-internal-li
34. Rust for Embedded Systems: Current State and Open Problems ... https://arxiv.org/html/2311.05063v2
35. Changelog 2026 | ClickHouse Docs https://clickhouse.com/docs/whats-new/changelog
36. B.2 The cc Options - Oracle Solaris Studio 12.2: C User's Guide https://docs.oracle.com/cd/E18659_01/html/821-1384/bjapr.html

37. [PDF] XL C: Compiler Reference - IBM https://www.ibm.com/docs/SSGH2K_13.1.3/com.ibm.compilers.aix.doc/compiler.pdf
38. gcc - GNU project C and C++ compiler - Ubuntu Manpage https://manpages.ubuntu.com/manpages/jammy/man1/alpha-linux-gnu-gcc.1.html
39. [PDF] Cisco Prime Network Registrar 11.1 Open Source Documentation https://www.cisco.com/c/dam/en/us/td/docs/net_mgmt/prime/network_registrar/11-1/third_party/open_source/Cisco_Prime_Network_Registrar_11_1_Open_Source_Document.pdf
40. Bringing the WebAssembly Standard up to Speed with SpecTec https://dl.acm.org/doi/full/10.1145/3656440
41. Building AI Coding Agents for the Terminal: Scaffolding, Harness ... https://arxiv.org/html/2603.05344v1
42. How to cross compile Rust across operating systems and CPU ... https://stackoverflow.com/questions/73642596/how-to-cross-compile-rust-across-operating-systems-and-cpu-architectures
43. GitHub pages are not updating [closed] - Stack Overflow https://stackoverflow.com/questions/20422279/github-pages-are-not-updating
44. Optimized third-party script loading for better web performance https://www.linkedin.com/posts/yomeshgupta_how-i-saved-18kb-js-bundle-with-one-simple-activity-7361615074236489731-yifu
45. GitHub | Kevin Bonett - LinkedIn https://www.linkedin.com/posts/kevbonett_github-basherweb-ui-boilerplate-accessible-activity-7262741193782870016-dfzA
46. 25 Developer Tools I Wish I Knew When I Started Coding https://dev.to/thebitforge/25-developer-tools-i-wish-i-knew-when-i-started-coding-1no0
47. A Novel Zero-Trust Identity Framework for Agentic AI - arXiv https://arxiv.org/html/2505.19301v1
48. Claude Code Simplifier Open Sourced: AI Agents Building Better Code https://www.linkedin.com/posts/kirankancheti_ai-machinelearning-developertools-activity-7416509449479180288-OSOl
49. [PDF] Red Hat Enterprise Linux 10 10.0 Release Notes https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/10/pdf/10.0_release_notes/Red_Hat_Enterprise_Linux-10-10.0_Release_Notes-en-US.pdf
50. An Empirical Analysis of Static Analysis Methods for Detection and ... https://arxiv.org/html/2604.07755v1
51. Static Code Analysis Rules - Technical Documentation For IFS Cloud https://docs.ifs.com/techdocs/25r2/060_development/050_development_tools/002_developer_studio/030_reference/810_static_code_rules/

52. LLM-Powered Static Binary Taint Analysis - ACM Digital Library https://dl.acm.org/doi/10.1145/3711816
53. How do I compile a Rust project to Wasm without using wasm-pack? https://stackoverflow.com/questions/60980310/how-do-i-compile-a-rust-project-to-wasm-without-using-wasm-pack
54. Goodbye Typescript, hello native typing for Javascript - DEV ... https://dev.to/christopherkade/goodbye-typescript-hello-native-typing-for-javascript-3ee1
55. Generating browser application bundles (phase: sealing) very slow https://stackoverflow.com/questions/67733861/angular-12-generating-browser-application-bundles-phase-sealing-very-slow
56. Passing LLM Instructions to Coders with Markdown Code Boxes https://www.linkedin.com/posts/piersfawkes_a-friday-ai-tip-when-im-passing-an-instruction-activity-7433163503865114624-p2VN
57. The AI Agent Code of Conduct: Automated Guardrail Policy-as ... https://arxiv.org/html/2509.23994v1
58. [PDF] Towards Enforcing Company Policy Adherence in Agentic Workflows https://aclanthology.org/2025.emnlp-industry.41.pdf
59. [PDF] Incentivizing Reasoning for Advanced Instruction-Following of Large ... https://openreview.net/pdf/0de94b848e4b6f73ac84843d481395c51d499376.pdf
60. Leveraging large language models for literature-driven prioritization ... https://academic.oup.com/bioinformatics/article/41/8/btaf449/8225722
61. A Scalable Framework for Multi-Policy AI Compliance Evaluation https://arxiv.org/html/2601.11702v1
62. Quality of Service Configuration Guide, Cisco IOS XE 17.17.x ... https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9500/software/release/17-17/configuration_guide/qos/b_1717_qos_9500_cg/configuring_qos.html
63. Configure a Login Banner for the UI Login Screen of the VMware ... https://techdocs.broadcom.com/us/en/vmware-cis/vsphere/vsphere/8-0/vsphere-single-host-management-vmware-host-client/vmware-host-client-overview-vSphereSingleHostManagementVMwareHostClient/using-the-vsphere-host-client-vSphereSingleHostManagementVMwareHostClient/configure-a-security-login-banner-for-the-ui-login-screen-vSphereSingleHostManagementVMwareHostClient.html
64. The New Separation of Powers - A Theory For The Modern State ... https://www.scribd.com/document/569030780/The-new-separation-of-powers-a-theory-for-the-modern-state-PDFDrive
65. 深入浅出聊聊Rust WebAssembly（一） - 稀土掘金 https://juejin.cn/post/6971986364069855239

66. WebAssembly Memory Tagging | Proceedings of the 2025 Cloud ... https://dl.acm.org/doi/10.1145/3733812.3765536
67. SentinelAgent: Intent-Verified Delegation Chains for Securing ... - arXiv https://arxiv.org/html/2604.02767v1
68. Smart Contracts as Semantic Signalling for AI Agents on Blockchains https://dl.acm.org/doi/10.1145/3793638.3793651
69. Arxiv今日论文| 2026-04-13 - 闲记算法 http://lonepatient.top/2026/04/13/arxiv_papers_2026-04-13.html
70. Context Engineering for Commercial Agent Systems - Jeremy Daly https://www.jeremydaly.com/context-engineering-for-commercial-agent-systems/
71. Compliance-as-Code for AI-Driven Identity Systems - IEEE Xplore https://ieeexplore.ieee.org/iel8/6287639/11323511/11398064.pdf
72. [PDF] SCEaR Newsletter Special IssueFinalFinal - UNESCO https://www.unesco.org/sites/default/files/medias/fichiers/2025/10/SCEar_Special_Issue_2024_Artificial_Intelligence_and_Documentary_Heritage.pdf
73. Failed to parse source map - javascript - Stack Overflow https://stackoverflow.com/questions/70599784/failed-to-parse-source-map
74. Metadata Vocabulary for Tabular Data - W3C https://www.w3.org/TR/2015/WD-tabular-metadata-20150108/
75. HelpSteer3-Preference: Open Human-Annotated Preference Data ... https://arxiv.org/html/2505.11475v1
76. An Analysis of Modern Web Security Vulnerabilities Inside ... - arXiv https://arxiv.org/html/2603.09426v1
77. Deep Dive: Why Rust-Based Tooling is Dominating JavaScript in 2026 https://dev.to/dataformathub/deep-dive-why-rust-based-tooling-is-dominating-javascript-in-2026-3dbl
78. PASTA: A Scalable Framework for Multi-Policy AI Compliance ... https://dl.acm.org/doi/10.1145/3772318.3790966
79. [PDF] Policy-as-Prompt: Turning AI Governance Rules into Guardrails for ... https://openreview.net/pdf?id=8TMSomzq6y
80. [PDF] Ontology-to-tools compilation for executable semantic constraint ... https://arxiv.org/pdf/2602.03439
81. (PDF) PASTA: A Scalable Framework for Multi-Policy AI Compliance ... https://www.researchgate.net/publication/399931776_PASTA_A_Scalable_Framework_for_Multi-Policy_AI_Compliance_Evaluation

82. A Formal Specification Smart-Contract Language for Legally Binding ... https://ieeexplore.ieee.org/iel7/6287639/9312710/09435331.pdf
83. [PDF] The 10th Workshop on Financial Technology and Natural Language ... https://aclanthology.org/2025.finnlp-2.pdf
84. [PDF] SoK: Agentic Skills -Beyond Tool Use in LLM Agents | HAL https://hal.science/hal-05525481v1/file/_Sok_Agent_Skills.pdf
85. [PDF] Sovereign Agents: Towards Infrastructural Sovereignty and Diffused ... https://arxiv.org/pdf/2602.14951
86. [PDF] Constrained Reinforcement for Bounded Decision Authority - arXiv https://arxiv.org/pdf/2603.02019
87. The New Sovereignty in International Relations - jstor https://www.jstor.org/stable/3186572
88. [PDF] The concept of class invariant in object-oriented programming - arXiv https://arxiv.org/pdf/2109.06557
89. PBFuzz: Agentic Directed Fuzzing for PoV Generation - arXiv https://arxiv.org/html/2512.04611v1
90. How to overcome the regex depth limitation in python? https://stackoverflow.com/questions/53618278/how-to-overcome-the-regex-depth-limitation-in-python
91. User Guide for AsyncOS 13.5.1 for Cisco Email Security Appliances https://www.cisco.com/c/en/us/td/docs/security/esa/esa13-5-1/user_guide/b_ESA_Admin_Guide_13-5-1/b_ESA_Admin_Guide_12_1_chapter_01000.html
92. Syntax Is Easy, Semantics Is Hard: Evaluating LLMs for LTL ... - arXiv https://arxiv.org/html/2604.07321v1
93. User Guide — Nsight Systems - NVIDIA Documentation Hub https://docs.nvidia.com/nsight-systems/UserGuide/index.html
94. 虚幻引擎5.1版本说明 https://dev.epicgames.com/documentation/zh-cn/unreal-engine/unreal-engine-5.1-release-notes%3Fapplication_version%3D5.1?application_version=5.1
95. Unity 2021.2.0f1 https://unity.com/releases/editor/whats-new/2021.2.0f1
96. Different behavior for same wasm file between local instantiation ... https://stackoverflow.com/questions/74145481/different-behavior-for-same-wasm-file-between-local-instantiation-and-github-act
97. A Survey of Vibe Coding with Large Language Models - arXiv https://arxiv.org/html/2510.12399v2
98. About GitHub Copilot cloud agent https://docs.github.com/copilot/concepts/agents/coding-agent/about-coding-agent

99. The Auton Agentic AI Framework A Declarative Architecture for ... https://arxiv.org/html/2602.23720v1
100. A Security Governance Specification for AI Agent Tool Access - arXiv https://arxiv.org/html/2602.20196v1
101. AI Skills as the Institutional Knowledge Primitive for Agentic Software ... https://arxiv.org/html/2603.14805v1
102. Qualixar OS: A Universal Operating System for AI Agent Orchestration https://arxiv.org/html/2604.06392v1
103. Talk Freely, Execute Strictly: Schema-Gated Agentic AI for Flexible ... https://arxiv.org/html/2603.06394v1
104. Autonomous Agents on Blockchains: Standards, Execution Models ... https://arxiv.org/html/2601.04583v1
105. ParseBench: A Document Parsing Benchmark for AI Agents - arXiv https://arxiv.org/html/2604.08538v3
106. [PDF] OpenPort Protocol: A Security Governance Specification for AI Agent ... https://www.arxiv.org/pdf/2602.20196
107. Code Digital Twin: A Knowledge Infrastructure for AI-Assisted ... - arXiv https://arxiv.org/html/2503.07967v4
108. [PDF] the law of cyber-space - Biblioteca Cejamericas https://biblioteca.cejamericas.org/bitstream/handle/2015/3576/Law_Cyber_Space_Negotiations.pdf?sequence=1&isAllowed=y
109. Module parse failed for wasm module generated by rust wasm-pack https://stackoverflow.com/questions/72222735/module-parse-failed-for-wasm-module-generated-by-rust-wasm-pack
110. How to load library source maps using webpack? - Stack Overflow https://stackoverflow.com/questions/32702478/how-to-load-library-source-maps-using-webpack
111. Using Typescript without compilation - DEV Community https://dev.to/thepassle/using-typescript-without-compilation-3ko4
112. MDEval: Evaluating and Enhancing Markdown Awareness in Large ... https://dl.acm.org/doi/abs/10.1145/3696410.3714674
113. zoom-oss-attribution-june-2025.txt https://www.zoom.com/siteassets/opensource/en/2025/zoom-oss-attribution-june-2025.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669

114. Schema-Governed LLM Pipeline for Executable Narrative ... - MDPI https://www.mdpi.com/2079-8954/14/2/175
115. [PDF] Hierarchical Attention Based Convolutional Graph Neural Network ... https://aclanthology.org/2021.findings-emnlp.381.pdf
116. Guide to Writing Effective Copilot Studio Agent Instructions - LinkedIn https://www.linkedin.com/pulse/guide-writing-effective-copilot-studio-agent-pierre-yves-delac%C3%B4te-sdcye
117. Permission Manifests for Web Agents Lightweight Agent Standards ... https://arxiv.org/html/2601.02371v1
118. 转载| Harness Engineering: 基于Claude Code 的完全指南 - 博客园 https://www.cnblogs.com/zerofix/p/19849814
119. [PDF] On the Optimal Design of a Financial Stability Fund https://www.liuyanecon.com/wp-content/uploads/RES2025-FSF.pdf
120. Sovereign Debt as a Contingent Claim: Excusable Default ... - jstor https://www.jstor.org/stable/1807168
121. [PDF] Sovereign bond-backed securities: a feasibility study – Volume II https://www.esrb.europa.eu/pub/task_force_safe_assets/shared/pdf/esrb.report290118_sbbs_volume_II_technicalanalysis.en.pdf
122. A Multi-Agent Framework for Automated Exploit Generation ... - arXiv https://arxiv.org/html/2604.05130
123. Unity Editor Beta Releases https://unity.com/cn/releases/editor/beta
124. Terms query | Elasticsearch Reference https://www.elastic.co/docs/reference/query-languages/query-dsl/query-dsl-terms-query
125. Azure Application Gateway v1 - Azure updates | Microsoft Azure https://azure.microsoft.com/updates/application-gateway-v1-will-be-retired-on-28-april-2026-transition-to-application-gateway-v2/
126. Artificial Intelligence - arXiv https://arxiv.org/list/cs.AI/new
127. Problem with migrated rails 8 sprockets/importmap ... - Stack Overflow https://stackoverflow.com/questions/79819553/problem-with-migrated-rails-8-sprockets-importmap-sprockets-errors-on-app-javasc
128. Faster script loading with BinaryAST? - The Cloudflare Blog https://blog.cloudflare.com/binary-ast/
129. abstract-syntax-tree - NPM https://www.npmjs.com/package/abstract-syntax-tree
130. How to solve "#error Unsupported architecture" when building Rust ... https://stackoverflow.com/questions/55792152/how-to-solve-error-unsupported-architecture-when-building-rust-code-with-wasm

131. Webassembly-学习手册-全- - 绝不原创的飞龙- 博客园 https://www.cnblogs.com/apachecn/p/19164929
132. Towards Repo-Level Generation via Structured Symbolic Paradigm https://arxiv.org/html/2604.13100v1
133. International Joint Conference on Natural Language Processing ... https://aclanthology.org/events/ijcnlp-2019/
134. Arxiv今日论文| 2026-01-15 - 闲记算法 http://lonepatient.top/2026/01/15/arxiv_papers_2026-01-15
135. Reasoning-agent-driven process simulation, optimization, carbon ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12891574/
136. zoom-oss-attribution-october-2025.txt https://www.zoom.com/siteassets/opensource/en/2025/zoom-oss-attribution-october-2025.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669
137. InCoder-32B: Code Foundation Model for Industrial Scenarios - arXiv https://arxiv.org/html/2603.16790v1
138. SEAL: Integrating Program Analysis and Repository Mining https://dl.acm.org/doi/10.1145/3585008
139. Advancing Large Language Models for Code Using Code-Structure ... https://escholarship.org/uc/item/94x2m8zp
140. Use of regex in lexical analysis - Stack Overflow https://stackoverflow.com/questions/16880736/use-of-regex-in-lexical-analysis
141. APOLLO: Automated LLM and Lean Collaboration for Advanced ... https://arxiv.org/html/2505.05758v5
142. [PDF] Release 0.23.0 Matthias Valvekens - pyHanko https://pyhanko.readthedocs.io/_/downloads/en/v0.23.0/pdf/
143. CEF audit logs | TrendAI™ - Trend Micro Online Help Center https://docs.trendmicro.com/en-us/documentation/article/trend-vision-one-cef-audit-logs
144. AgentCity: Constitutional Governance for Autonomous Agent ... - arXiv https://arxiv.org/html/2604.07007v1
145. GLP: A Grassroots, Multiagent, Concurrent, Logic Programming ... https://arxiv.org/html/2510.15747v2
146. [PDF] Autonomous Agents on Blockchains: Standards, Execution Models ... https://arxiv.org/pdf/2601.04583
147. Computer Science - arXiv https://arxiv.org/list/cs/new

148. [PDF] A Grassroots, Multiagent, Concurrent, Logic Programming Language https://arxiv.org/pdf/2510.15747
149. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=25&show=2000
150. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=200&show=2000
151. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=600&show=2000
152. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=500&show=500
153. A Survey of Bugs in AI-Generated Code - arXiv https://arxiv.org/html/2512.05239v1
154. Unboxed Data Constructors: Or, How cpp Decides a Halting Problem https://dl.acm.org/doi/10.1145/3632893
155. Packages in need of a new maintainer - Debian https://www.debian.org/devel/wnpp/work_needing
156. Performance and interpretability analysis of code generation large ... https://www.sciencedirect.com/science/article/pii/S0925231225021332
157. Domain Specific Language for Smart Contract Development https://www.researchgate.net/publication/343707820_Domain_Specific_Language_for_Smart_Contract_Development
158. zoom-oss-attribution-december-2025.txt https://www.zoom.com/siteassets/opensource/en/2025/zoom-oss-attribution-december-2025.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669
159. Considerations in adopting RHEL 9 | Red Hat Enterprise Linux https://docs.redhat.com/pt-br/documentation/red_hat_enterprise_linux/9/html-single/considerations_in_adopting_rhel_9/index
160. Web World Models - arXiv https://arxiv.org/html/2512.23676v1
161. 更新日志— Python 3.14.4 文档 https://docs.python.org/zh-cn/3/whatsnew/changelog.html
162. [PDF] OpenShift Container Platform 4.21 Service Mesh https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/pdf/service_mesh/OpenShift_Container_Platform-4.21-Service_Mesh-en-US.pdf
163. Directory: /homebrew-bottles/api/formula - NJU Mirror https://mirror.nju.edu.cn/homebrew-bottles/api/formula/
164. Can I handle imports in an Abstract Syntax Tree? - Stack Overflow https://stackoverflow.com/questions/54325116/can-i-handle-imports-in-an-abstract-syntax-tree
165. astring - NPM https://www.npmjs.com/package/astring

166. The 2024 Joint International Conference on Computational ... https://aclanthology.org/events/lrec-2024/
167. Arxiv今日论文| 2026-02-25 - 闲记算法 http://lonepatient.top/2026/02/25/arxiv_papers_2026-02-25.html
168. A Survey of Practical Formal Methods for Security - ACM Digital Library https://dl.acm.org/doi/full/10.1145/3522582
169. Computer Science - arXiv https://www.arxiv.org/list/cs/new?skip=25&show=1000
170. props.conf - Splunk Documentation https://docs.splunk.com/Documentation/Splunk/8.2.12/Admin/Propsconf
171. Query string query | Elasticsearch Reference https://www.elastic.co/docs/reference/query-languages/query-dsl/query-dsl-query-string-query
172. Cisco Nexus Dashboard Insights Analysis Hub, Release 6.5.x https://www.cisco.com/c/en/us/td/docs/dcn/ndi/6x/articles-651/ndi-analysis-hub-aci.html
173. vega-embed - NPM https://www.npmjs.com/package/vega-embed
174. [2206.01962] Formal Specifications from Natural Language - arXiv https://arxiv.org/abs/2206.01962
175. Adding repository custom instructions for GitHub Copilot https://docs.github.com/copilot/customizing-copilot/adding-custom-instructions-for-github-copilot
176. Adding custom instructions for GitHub Copilot CLI https://docs.github.com/en/copilot/how-tos/copilot-cli/customize-copilot/add-custom-instructions
177. Evaluating Large Language Models as IDE Agents on Real-World ... https://arxiv.org/html/2601.20886v1
178. [PDF] Demystifying Invariant Effectiveness for Securing Smart Contracts ... https://arxiv.org/pdf/2404.14580
179. [PDF] Verification of smart contracts: A survey - HAL https://hal.science/hal-03492408/document
180. Why are these regular expressions not working? - Stack Overflow https://stackoverflow.com/questions/44378607/why-are-these-regular-expressions-not-working
181. Firewall Rule Regex Not Working - SSL / TLS - Cloudflare Community https://community.cloudflare.com/t/firewall-rule-regex-not-working/173630
182. The role of metadata in reproducible computational research - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8441584/
183. [PDF] arXiv:2110.08362v3 [cs.PL] 13 Feb 2022 https://arxiv.org/pdf/2110.08362
184. LSPRAG: LSP-Guided RAG for Language-Agnostic Real-Time Unit ... https://arxiv.org/html/2510.22210v2

185. License info - DHL https://www.dhl.com/fragments/gcr-v1/static/media/licenses.a079219693c45e7c8635.txt

186. zoom-oss-attribution---march-2021.txt https://www.zoom.com/siteassets/opensource/en/2021/zoom-oss-attribution---march-2021.txt?filter-products=&filter-countries=&keywords=?pageSize=9&+Consulting=&page=1&sort=newest&categories=security-privacy&optimizely_user_id=a916f6f81cd61f75eb1d87620d9ab0df&onlycontent=1&DeviceId=24955141-ac84-4d3d-b61f-dc48472c8b12&SessionId=1769817600669

187. Open Data Kit User Guide | PDF | Qr Code | Google Play - Scribd https://www.scribd.com/document/643858510/ODK-Documentation

188. Artificial Intelligence and Machine Learning for Digital Pathology https://link.springer.com/content/pdf/10.1007/978-3-030-50402-1.pdf

189. [PDF] Verifying Declarative Smart Contracts https://faculty.sist.shanghaitech.edu.cn/hxchen/papers/2024-icse-dcv.pdf

190. Demystifying Invariant Effectiveness for Securing Smart Contracts ... https://arxiv.org/html/2404.14580v2

191. Demystifying Invariant Effectiveness for Securing Smart Contracts https://dl.acm.org/doi/full/10.1145/3660786

192. [PDF] Verification of smart contracts: A survey - ScienceDirect.com https://www.sciencedirect.com/science/article/am/pii/S1574119220300821

193. Contract-Graph Fusion and Cross-Graph Matching for Smart ... - MDPI https://www.mdpi.com/2076-3417/15/19/10844

194. Strengthening Solidity Invariant Generation: From Post - arXiv https://arxiv.org/html/2409.01804v2

195. (PDF) Strengthening Solidity Invariant Generation: From Post- to Pre ... https://www.researchgate.net/publication/383744783_Strengthening_Solidity_Invariant_Generation_From_Post-_to_Pre-Deployment

196. Automated Invariant Generation For Solidity Smart Contracts-2 - Scribd https://www.scribd.com/document/967738752/Automated-Invariant-Generation-for-Solidity-Smart-Contracts-2